

Session 2 Quiz - AI Concepts & Terminology

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:56582]

AI that creates brand-new things like text, images, or music is called _____ AI.

- A. Deep
- B. Predictive
- C. Generative
- D. General

Correct Answer: C. Generative

Explanation: Generative AI is specifically named for its ability to generate or create new content such as text, images, music, or video. Unlike AI that only analyzes or classifies existing data, generative AI produces something that did not exist before. Deep, Predictive, and General refer to other AI concepts that do not focus on creating new content.

Q2 [Type: MCQ | Difficulty: Easy | QID:56583]

A(n) _____ is like a recipe that provides step-by-step instructions for a computer to follow.

- A. Algorithm
- B. Model
- C. Database
- D. Sensor

Correct Answer: A. Algorithm

Explanation: An algorithm is a precise sequence of steps or rules a computer follows to solve a problem or complete a task, just like a cooking recipe gives step-by-step directions. A model is the result after training, a database stores data, and a sensor collects information from the environment.

Q3 [Type: MCQ | Difficulty: Easy | QID:56584]

The saying 'Garbage In, _____' means that if an AI is trained on bad data, it will produce bad results.

- A. Nothing Out
- B. Data Out
- C. Machine Out
- D. Garbage Out

Correct Answer: D. Garbage Out

Explanation: "Garbage In, Garbage Out" (GIGO) is a well known principle in computing and AI. It means that the quality of the output is directly determined by the quality of the input. If the training data is flawed, incomplete, or full of errors, the AI will produce flawed results no matter how good the algorithm is.

Q4 [Type: MCQ | Difficulty: Easy | QID:56585]

The process where an AI studies millions of examples to find patterns is called _____.

- A. Coding
- B. Training
- C. Predicting
- D. Sensing

Correct Answer: B. Training

Explanation: Training is the phase where an AI algorithm processes large amounts of data, learns patterns, and improves its performance. Just as athletes train to get better at a sport, AI systems train on data to become better at their tasks. Coding is writing instructions, predicting is making an output, and sensing is collecting data.

Q5 [Type: MCQ | Difficulty: Easy | QID:56586]

Information like photos, temperatures, and text messages used by AI is collectively called _____.

- A. Models
- B. Hardware
- C. Data
- D. Predictions

Correct Answer: C. Data

Explanation: Data is the broad term for all raw information that AI systems use to learn and function. Photos, temperatures, text messages, videos, and sensor readings are all examples of data. Hardware refers to physical components, models are the result of training, and predictions are the outputs.

Q6 [Type: True/False | Difficulty: Easy | QID:56587]

True or False: A 'Model' is the finished, ready-to-use version of an AI that has completed its training.

- A. True
- B. False

Correct Answer: A. True

Explanation: After training is complete, the AI's learned patterns and knowledge are saved as a model. This model can then be used to make predictions or decisions on new data. The model is indeed the finished, ready-to-use result of the training process.

Q7 [Type: True/False | Difficulty: Easy | QID:56588]

True or False: Data from a smartwatch tracking your heart rate at 72 bpm is an example of sensor data.

A. True

B. False

Correct Answer: A. True

Explanation: Smartwatches use physical sensors (optical heart rate sensors) to measure your pulse. Any data collected by a physical device that measures real world conditions (temperature, motion, heart rate, etc.) is called sensor data. The heart rate reading of 72 bpm fits this definition perfectly.

Q8 [Type: True/False | Difficulty: Easy | QID:56589]

True or False: Every word generated by an AI model like Gemini is a prediction based on patterns it learned during training.

A. False

B. True

Correct Answer: B. True

Explanation: Large language models like Gemini do not understand meaning. They work by predicting the next most likely word based on statistical patterns learned from millions of text examples. Each word they generate is essentially a calculated guess, not a thought or understanding.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:56590]

True or False: An algorithm and a model are the same thing because they both use data to work.

- A. True
- B. False

Correct Answer: B. False

Explanation: An algorithm is the set of instructions or recipe used before learning takes place. A model is the specific result after the algorithm has processed the training data. Think of it this way: the algorithm is the recipe, and the model is the finished cake. They are not the same thing.

Q10 [Type: True/False | Difficulty: Easy | QID:56591]

True or False: If you only train an AI with photos of brown dogs, it will still easily recognize a white dog.

- A. False
- B. True

Correct Answer: A. False

Explanation: AI can only recognize patterns that were present in its training data. If an AI has only seen brown dogs, it may learn that "brown" is a necessary feature of being a dog. When shown a white dog, it may fail to recognize it as a dog. This is an example of "Garbage In, Garbage Out" and shows why diverse data is important.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:56592]

Which term describes the AI's 'best guess' when it encounters something new after its training is complete?

- A. Prediction
- B. Data Collection
- C. Flashcarding
- D. Instruction

Correct Answer: A. Prediction

Explanation: In AI, a prediction is the output or guess that a trained model makes when given new, unseen data. For example, when you show a trained cat classifier a new photo, its "best guess" of whether it is a cat is called a prediction. Data collection, flashcarding, and instruction are not standard terms for this output.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:56593]

According to the source material, how is an algorithm similar to a cooking recipe?

- A. Both are only used for making food-related AI
- B. Both use specific steps to reach a final result
- C. Both require a computer to be completed
- D. Both are always 100% accurate every time

Correct Answer: B. Both use specific steps to reach a final result

Explanation: The core similarity is that both an algorithm and a cooking recipe provide a logical, step by step sequence of actions to transform inputs (data or ingredients) into a desired output (a model or a finished dish). The other options are incorrect: algorithms are not food related, recipes do not require computers, and neither is always 100% accurate.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:56594]

Which of these everyday tools represents a 'Model' that has been trained on a specific person's data?

- A. Face Unlock on a phone
- B. A standard calculator
- C. A physical wall clock
- D. A blank notebook

Correct Answer: A. Face Unlock on a phone

Explanation: Face Unlock is an AI model that has been specifically trained on your face. The phone learned your unique facial features during setup. A calculator follows fixed rules (not trained), a wall clock has no AI, and a blank notebook contains no data or training.

Q14 [Type: MCQ | Difficulty: Medium | QID:56595]

What specifically happens during the 'Training' phase of the AI Recipe?

- A. The AI makes its very first correct prediction
- B. A human writes down their password
- C. The computer is turned off to save energy
- D. The algorithm processes data to create a model

Correct Answer: D. The algorithm processes data to create a model

Explanation: Training is the middle step where the algorithm (instructions) acts upon the data (examples) to build the final model (the learned knowledge). This is the core learning phase of AI. The other options describe unrelated or incorrect events.

Q15 [Type: MCQ | Difficulty: Medium | QID:56596]

Which of the following is the correct order for the 'AI Recipe' discussed in class?

- A. Data + Algorithm → Training → Model → Prediction
- B. Algorithm → Prediction → Data → Training
- C. Prediction → Model → Training → Data
- D. Training → Data → Algorithm → Model

Correct Answer: A. Data + Algorithm → Training → Model → Prediction

Explanation: The correct sequence starts with data and an algorithm as ingredients. These go into training, which produces a model. Finally, the model is used to make predictions on new inputs. This flow is fundamental to understanding how modern AI systems are built and used.

Q16 [Type: MCQ | Difficulty: Medium | QID:56597]

If two different companies use the same AI algorithm but different data, why do they end up with different models?

- A. Computers have unique personalities like humans
- B. The model depends on the specific patterns in the data it saw
- C. One company is probably following the instructions wrong
- D. Algorithms change their own rules automatically

Correct Answer: B. The model depends on the specific patterns in the data it saw

Explanation: Training an AI is like studying for a test. Even if two students use the same study method (algorithm), they will learn different things if they study different materials (data). The resulting model reflects the specific patterns, examples, and biases present in its training data.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:56598]

In the 'Flashcard Story' analogy, what specifically do the cards with pictures represent?

- A. The Prediction
- B. The Model
- C. The Algorithm
- D. The Data

Correct Answer: D. The Data

Explanation: In the flashcard analogy, the cards with pictures are the examples and information that the learner studies. That is exactly what data is for an AI system: the raw examples (pictures, text, numbers) from which it learns. Predictions, models, and algorithms are different parts of the process.

Q18 [Type: MCQ | Difficulty: Medium | QID:56599]

Looking at the 'Big Picture' of AI, which of the following categories is the MOST specific (the smallest circle)?

- A. Deep Learning (DL)
- B. Generative AI (GenAI)
- C. Machine Learning (ML)
- D. Artificial Intelligence (AI)

Correct Answer: B. Generative AI (GenAI)

Explanation: The hierarchy from largest to smallest is: Artificial Intelligence (broadest), then Machine Learning (a subset of AI), then Deep Learning (a subset of ML), then Generative AI (a specific type of deep learning that creates new content). GenAI is therefore the most specific or smallest circle.

Q19 [Type: MCQ | Difficulty: Easy | QID:56600]

A text message saying 'Hi!' is categorized as which type of data?

- A. Numbers
- B. Videos
- C. Sensors
- D. Words

Correct Answer: D. Words

Explanation: A text message consists of characters, letters, and language. This falls into the category of "words" or text data. It is not numeric data, not a video, and not sensor data (which comes from physical measurement devices).

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:56601]

Which of the following best explains why people who build AI must think carefully about WHAT data they use?

- A. The AI only knows what it has seen in its training data
- B. Data is very expensive to store on a phone
- C. The AI will get bored if the data is all the same
- D. Using too much data makes the computer break

Correct Answer: A. The AI only knows what it has seen in its training data

Explanation: AI systems cannot reason beyond the patterns present in their training data. If the data is missing certain examples (like white dogs when training only on brown dogs), the AI will be limited or wrong. This is why data quality, diversity, and completeness are critical. The other options are false or misleading.